

Program-5 : Demonstrate CI Integration by Running Jenkins in a Docker Container

Description :

This program demonstrates Continuous Integration (CI) by deploying and running Jenkins inside a Docker container using Docker Compose. It shows how Docker simplifies Jenkins installation and enables automated build workflows.

Step-1 : Verify Docker and Docker Compose

```
docker --version
docker compose version
```

Step-2 : Create Project Directory

```
cd ~
mkdir program-5
cd program-5
```

Step-3 : Create Dockerfile

```
nano Dockerfile

FROM jenkins/jenkins:its
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Step-4 : Create docker-compose.yml

```
nano docker-compose.yml

version: "3.8"
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step-5 : Build and Run Jenkins using Docker Compose

```
sudo docker compose up --build
```

```

pavan-k@pavank1RV24MC072:~$ mkdir program-5
cd program-5
pavan-k@pavank1RV24MC072:~/program-5$ nano Dockerfile
pavan-k@pavank1RV24MC072:~/program-5$ nano docker-compose.yml
pavan-k@pavank1RV24MC072:~/program-5$ sudo docker compose up --build
[sudo] password for pavan-k:
WARN[0000] /home/pavan-k/program-5/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 71.7s (8/8) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 506B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 136B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts 3.5s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 25.6s
=> => resolve docker.io/jenkins/jenkins:lts@sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7 0.0s
=> => sha256:d2519f76d4c6527e03084f2742da3ae15b8c5097d464be4dd93de4f573749313 392B / 392B 0.3s
=> => sha256:3fb70446889ebcc58e91f43f90aaf314798941c318ce016ea62060a3de53927 2.23kB / 2.23kB 0.6s

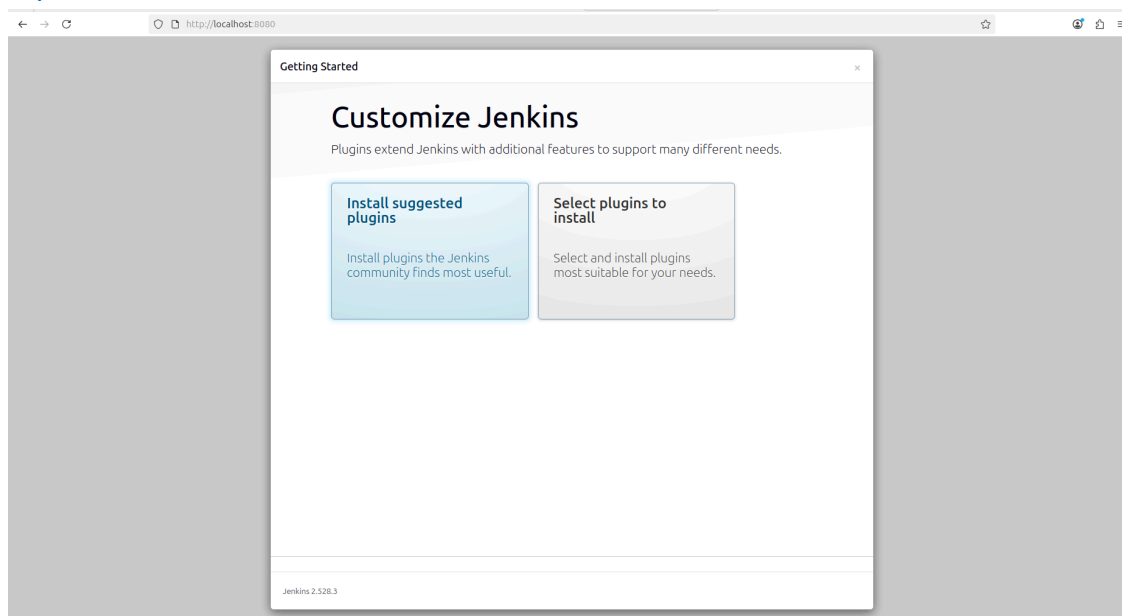
jenkins-ci | [LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
jenkins-ci | [LF]> Please use the following password to proceed to installation:
jenkins-ci | [LF]>
jenkins-ci | [LF]> e2206b9278864c31abaa7379af414e05
jenkins-ci | [LF]>
jenkins-ci | [LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
jenkins-ci | [LF]>
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci |
jenkins-ci | 2026-01-04 19:35:47.933+0000 [id=69] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
jenkins-ci | 2026-01-04 19:35:47.950+0000 [id=42] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
jenkins-ci | 2026-01-04 19:35:49.702+0000 [id=86] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
jenkins-ci | 2026-01-04 19:35:49.702+0000 [id=86] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1

```

w Enable Watch **d** Detach

Step-6 : Access Jenkins

<http://localhost:8080>



Step-7 : List Docker Images and Remove Jenkins Image

sudo docker images

sudo docker image rm program5-jenkins:latest

Result :

Jenkins is successfully deployed inside a Docker container using Docker Compose and Continuous Integration is demonstrated.